

Tutorial 1

1. Consider the following data on diastolic blood pressure (measured in mm Hg) for 9 patients. Use R to answer the questions below and leave your final answers to 2 decimal places.

Patient	1	2	3	4	5	6	7	8	9
Blood pressure	96	119	119	108	126	128	110	105	94

- (a) Input the data into R and sort it. Manually find the median of the dataset, then use R to get the median. Are the results the same?
 - (b) Determine the mean, variance and standard deviation of the dataset.
 - (c) Suppose we are now told that the measurement machine was not zero-ed, and all readings should be *reduced by 10 mm Hg*. What is the new mean and variance of the dataset? Does any of them change?
 - (d) Create a boxplot of the new data. Label the title and y-axis of your boxplot as accurately as possible.
 - (e) Create a histogram of the new data (using probability). Label the x-axis.
2. For each of the following experiments, identify the variable(s), the data type of each variable, and the sample size.
- (a) For each of 10 beetles, a biologist counted the number of times the beetle fed on a disease-resistant plant during a 4-hour period.
 - (b) In a nutritional study, 40 healthy males were measured for height and weight as well as the weight of their food intake over a 24-hour period.
 - (c) The birth weight, number of siblings, mother's race, and age were recorded for each of 85 babies born at NUH.
3. The histograms and boxplots of three datasets are plotted in Figure 1. Each dataset consists of 1000 points. Match the following phrases to each of the datasets: "symmetric and bimodal", "symmetric" and "skewed right".
4. The histograms and boxplots of three datasets are plotted below, in Figure 2. However, the order of plots in the top and bottom rows are not the same. Match the histogram with the correct boxplot.

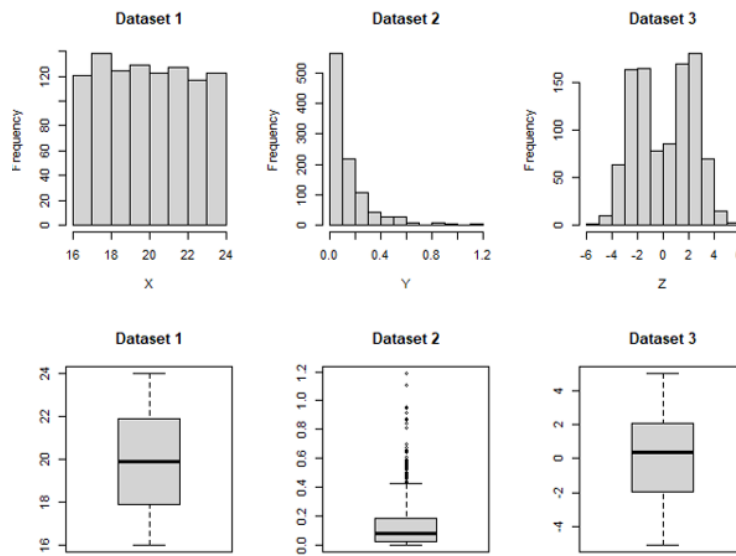


Figure 1: Match the phrases in the question to the plots.

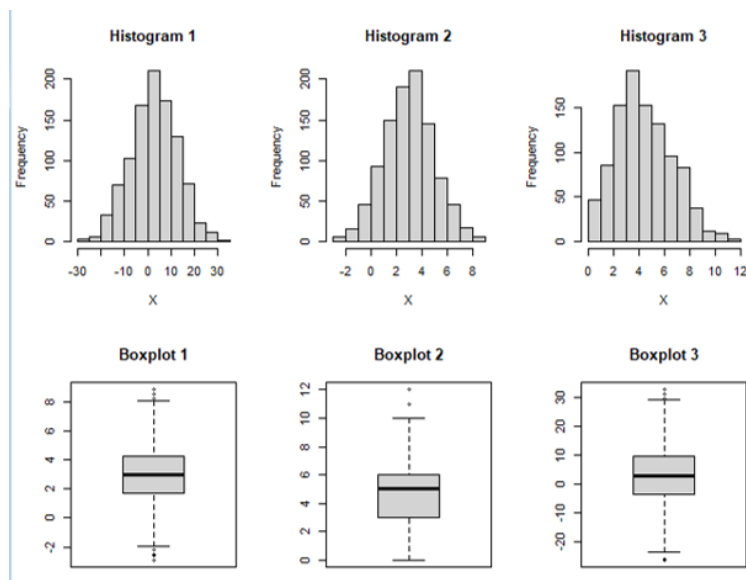


Figure 2: Match the histograms in the top row to the boxplots in the bottom row.